

Kyuwon Weon

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EDUCATION

Northwestern University | Evanston, IL
Master of Science in Robotics

December 2026 (expected)

Carnegie Mellon University (CMU) | Pittsburgh, PA
Bachelor of Science in Mechanical & Biomedical Engineering (University Honors)

Aug 2018 - May 2022

SKILLS

Robotics: ROS 2, MoveIt, TF2, OpenCV, Gazebo, MuJoCo, Coppeliasim, RViz, SLAM, Pinocchio, ProxSuite

Software: Python, C++, C, PyTorch, Linux (Ubuntu), Git, Unit Testing, MATLAB, Minitab

Control: State-Space Modeling, Lagrangian Dynamics, PID, Forward/Inverse Kinematics, Motion Planning, VLM

Mechanical: SolidWorks, Creo, ANSYS, COMSOL, GD&T, Design for Six Sigma, DFM, Rapid Prototyping

PROFESSIONAL EXPERIENCE

Alcon | *Medical Device Design Engineer* | Fort Worth, TX Jul 2022 - Jul 2025

- Led the full-lifecycle design of intraocular lens (IOL) implant devices, utilizing Creo and Six Sigma to scale from rapid prototyping to high-volume injection molding
- Engineered custom test fixtures for verification and validation testing under dynamic loading conditions
- Executed Root Cause Analysis using DOEs and Minitab statistical methods to resolve critical failure modes
- Collaborated within a multidisciplinary team and moderated clinical prototype evaluations with key opinion leaders (KOLs), translating feedback into design improvements

Alcon | *R&D Intern* | Belmont, CA Jun 2021 - Aug 2021

- Validated structural deformation predictions of fluid-accommodating IOL models via ANSYS
- Characterized the dynamic mechanical response of lens haptics under cyclic loading, analyzing stiffness and damping to predict long-term positional stability

PROJECTS/ RESEARCH

Robot Safety Control Architecture (CBF-QP) Jan - March 2026

- Architected a 1kHz real-time C++ safety filter, utilizing Control Barrier Functions (CBF) and Quadratic Programming (QP) to mathematically guarantee collision prevention for Franka manipulators in shared space
- Engineered Pinocchio-based kinematics pipeline in C++ with custom ROS 2 interface to handle whole-body constraints and dynamic obstacle interaction

EKF SLAM with TurtleBot3 Jan - March 2026

- Developed a full-stack SLAM pipeline from scratch in C++ with Extended Kalman Filter for TurtleBot3
- Extracted geometric landmarks from raw LiDAR point clouds via custom Euclidean clustering and circle-fitting, utilizing Mahalanobis distance for unknown data association

VLM-Driven Interactive Writing Robot Dec 2025

- Co-developed a VLM-based agent that perceives questions and writes answers using ROS 2 and Python
- Implemented motion planning in MoveIt to generate pen gripping and writing trajectories, while optimizing force/torque safety thresholds to permit writing on a dynamic, handheld whiteboard

Reactive Motion Planning & Stabilization Controller (APF) Dec 2025

- Implemented a reactive controller using Artificial Potential Fields in MuJoCo, ROS 2, and Python to navigate dynamic environments without pre-computed paths on a Franka Emika manipulator
- Engineered global damping algorithms to resolve local minima and high-frequency oscillations

Autonomous Pen Grabbing Manipulator Sep 2025

- Programmed a PincherX-100 arm to detect and grasp dynamic targets using OpenCV and RealSense